

Temporal-Support Identifiability in Event-Camera Detection Evaluation Pipelines

Anonymous CVPR submission

Paper ID *****

Abstract

001 *An event-camera detection benchmark identifies each*
002 *scored sample by one nominal timestamp, yet a recurrent*
003 *detector forms its prediction from a longer history whose*
004 *length the benchmark never records. We state the missing*
005 *requirement as temporal-support identifiability: the vari-*
006 *ables a protocol records with a score must uniquely de-*
007 *termine the temporal evidence behind every scored predic-*
008 *tion. The released SSM-ViT evaluation pipeline is the case*
009 *study: its reported score is not determined by the nomi-*
010 *nal timestamp alone, because it also depends on a temporal*
011 *support that the benchmark does not specify. With frames,*
012 *weights, labels, evaluator and evaluated timestamps fixed,*
013 *moving only the chunk boundary of the streaming evalua-*
014 *tion changes the recurrent history behind the same frames*
015 *from 1–4 to 17–20 windows, and changes the SSM-ViT*
016 *base checkpoint (S5-B) by 4.20 ± 0.36 mAP over all labeled*
017 *boxes and by 5.23 ± 0.49 mAP on the velocity-evaluable*
018 *subset; the released score pools histories of 1 to 21 win-*
019 *dows. Identifiability is a uniqueness property, so one con-*
020 *trolled pair with identical recorded variables and different*
021 *scores suffices to show that the protocol under-specifies its*
022 *score. The same intervention leaves RVT unchanged while*
023 *its state is carried and moves it by 5.39–6.91 mAP once the*
024 *state is reset, so the effect follows the reset policy of each*
025 *architecture. Support-Conditioned AP specifies the support:*
026 *every checkpoint is scored from the same reset state with the*
027 *history fixed to H windows. On five released Gen1 check-*
028 *points, S5-B is third in the pooled score and first at every*
029 *$H \in \{1, 5, 10, 21\}$, with every pairwise gap to it excluding*
030 *zero, and Kendall’s τ between the pooled and fixed- H or-*
031 *derings is at most 0.40. Predictor-side and label-side audits*
032 *show that input timing and label-acquisition timing are left*
033 *implicit by the same timestamp. We list five fields a bench-*
034 *mark can record with its labels so that the support behind a*
035 *score is identifiable.*

1. Introduction

Two recurrent detectors evaluated at the same nominal
timestamp are not comparable in time unless the history
each one actually used is also identified.

An event-camera detection benchmark defines a sample
by a nominal timestamp. A prediction $\hat{s}$ is scored against
the ground truth s^* carried at query time t through a func-
tion $m(\hat{s}, s^*)$ of position, shape and class. Beyond select-
ing the label at t , that function contains no variable for the
temporal evidence behind either argument, so any temporal
discrepancy enters the score disguised as displacement.

A recurrent detector does not read only the instant the
benchmark names. It computes $\hat{s}$ from an input window
and from state carried across earlier windows. The label it
is compared against comes from an exposure of finite width,
an annotator or tracker working on the exposed image, and
a geometric transfer into another sensor’s frame. The evi-
dence on both sides is an interval, while the protocol records
an instant.

We use three terms. The *nominal timestamp* is the in-
stant at which the protocol pairs a prediction with a label.
The *effective temporal support* is the past that can influence
the prediction, including state carried by recurrence. The
label-acquisition support is the interval over which the la-
bel’s evidence was acquired. The *protocol record* is the set
of variables a benchmark specifies and a paper reports with
a score: frames, labels, weights, evaluator, input window,
nominal timestamps and published acquisition metadata. A
protocol is *temporal-support identifiable* when its record
determines the effective temporal support of every scored
prediction uniquely. A quantity that can be reconstructed
from released code or index files, but is absent from the
record, does not meet this condition: two evaluations that
agree on the entire record can still differ in it, and nothing
reported with the score tells them apart.

We test it with a paired intervention on the released
SSM-ViT evaluation pipeline, using its base and small
checkpoints (S5-B, S5-S). In that implementation the nom-
inally carried state does not enter the outputs across eval-

uation chunks. The recurrent history available to a prediction is a deterministic function of chunk position, and ranges from 1 to 21 windows inside one pooled score. Moving every chunk boundary 5 windows later re-scores the same frames under a longer history at the same evaluated timestamp, with frames, weights, labels and evaluator fixed (3 574 labeled frames scored twice, cluster bootstrap over 406 sequences; $\pm$ denotes one cluster-bootstrap standard error).

Effective temporal support alone changes S5-B by 4.20 ± 0.36 mAP over all labeled boxes and 5.23 ± 0.49 mAP on the velocity-evaluable subset (Sec. 3). The second checkpoint of the same released pipeline, S5-S, moves by 4.10 and 4.81 points. Since temporal-support identifiability is a uniqueness property, this controlled pair is sufficient to show that the nominal timestamp does not uniquely specify the temporal evidence represented by that score. The consequence is at the level of the protocol, not of one implementation: where the protocol does not record the support behind a score, the reported score is underdetermined. The same intervention on RVT leaves it unchanged while its state is carried and moves it by 5.39 and 6.91 points once the state is reset at the boundary, so the effect belongs to the reset policy.

The controlled pair has a constructive counterpart. Support-Conditioned AP scores a checkpoint with the recurrent history fixed to a stated number of windows H (Sec. 5). Applied to five released Gen1 checkpoints on 19 946 common frames, it orders them differently from the pooled score: S5-B is third in the pooled score and first at each of $H = 1, 5, 10, 21$, and for three pairs the pooled and $H = 1$ gaps have intervals that exclude zero on opposite sides.

Contributions.

- *Temporal-support identifiability*: an evaluation property separating the nominal timestamp a protocol records from the effective temporal support and label-acquisition support a reported score represents.
- A paired intervention that isolates recurrent support. With frames, weights, labels, evaluator and evaluated timestamp fixed, only the chunk boundary changes, and the S5-B score moves by 4.20 mAP overall and 5.23 mAP on the velocity-evaluable subset. The same intervention on RVT follows the reset policy.
- A protocol, Support-Conditioned AP, which fixes the recurrent history behind each score, applied to five released checkpoints, together with five fields a benchmark can record so that the support behind a score is identifiable.

2. Related work

Sensor and runtime latency. Work on event-camera timing takes the benchmark’s temporal index as given and im-

proves what is measured against it. Sensor latency is calibrated away [22, 26], compute latency is folded into average precision [7, 27, 32], delay is scored separately [8], and temporal tolerance has been proposed [33]. Each changes what the predictor does at a fixed benchmark index. The object measured here is that index itself and the support a score reported under it pools: we test whether a released benchmark specification uniquely determines the temporal quantities behind its score.

Temporal windows and event benchmarks. RVT and SSM-ViT redesign or parameterize temporal windows [21, 23], and adaptive-window methods compare alternative partitions [25]. Gen1, 1 Mpx, DSEC-Det and Ev-3DOD use asynchronous measurements, rectified frames, or interpolated labels [14, 16–18], and the event count inside one labeled box can be too small to support detection [2]. Event-RGB frequency mismatch and exposure duration have also been noted [19, 20, 34]. LEOD [24] states that event detectors take only events triggered before the query time and builds a time-flip augmentation on that premise. It does not record where inside that window the evidence sits. Both released detector implementations leave a frame’s position inside the evaluation chunk unspecified by the protocol. The recurrent state entering a chunk is the state saved at the end of the previous chunk, and is reset only at the first sample of a sequence. Under streaming sampling every timestep emits predictions, while otherwise only the last does (RVT modules/detection.py:232, 243 and SSM-ViT :315, 318).

Benchmark reliability and evaluation analysis. Prior work studies detector calibration, stratified model evaluation, label-induced benchmark instability, and event-vision evaluation protocols [28–31]. These analyses examine uncertainty or error in reported performance, whereas our focus is the temporal support and acquisition semantics that a nominal detection timestamp leaves implicit.

Label time and attribution. Waymo and annotation studies estimate per-object capture or annotation offsets [3–5]. Event datasets expose per-box time fields, but their degrees of freedom are not necessarily specified. LET-3D-AP and TIDE/DETAD/HOTA decompose or bucket spatial error while taking the label clock as given [1, 9–12].

3. Primary evidence: the same-frame chunk-boundary intervention

The state the SSM-ViT release carries between evaluation chunks does not affect any of its outputs. It enters as a multiplicative factor on the first coefficient of an associative scan on which the outputs do not depend. Under the released

streaming evaluation this bounds the recurrent history available to a detection to the current 21-window chunk. That history ranges from 1 window to 21 with the detection's position in the chunk, and the reported number averages over all 21 positions. Chunk position is a deterministic function of the release's own index files and is recorded nowhere in the evaluation protocol.

The same-frame intervention. The intervention moves every chunk boundary 5 windows later and changes nothing else: the same frames, the same weights, the same labels, the same evaluator, and the evaluated timestamp of each frame unchanged. 3 574 labeled frames are scored twice under this shift, once with 1–4 windows of recurrent history and once with 17–20. They are compared paired within frame with a cluster bootstrap over the 406 sequences. The long-history arm gains 5.23 ± 0.49 mAP points for S5-B and 4.81 ± 0.52 for S5-S on the velocity-evaluable subset, and over every labeled box the S5-B gain is 4.20 ± 0.36 points. Mean detection confidence, which uses no labels, moves with it by 0.0579 ± 0.0048 and 0.0509 ± 0.0035 . Fig. 1 (bottom) draws one such pair. No quantity the protocol records distinguishes the two arms.

Graded response and what moves. The size of the boundary shift is a free parameter. Re-scoring the same 3 574 frames after shifting the boundary so that the starved block gains +6, +11 and +16 windows of history changes the velocity-subset mAP of S5-B by 3.73 ± 0.35 , 4.91 ± 0.50 and 5.23 ± 0.49 points, and that of S5-S by 3.54 ± 0.39 , 4.29 ± 0.52 and 4.81 ± 0.52 . The zero-shift control, which re-runs the release's own boundary placement, returns 0.00. Confidence on detections matched to a label rises by 0.0409 and 0.0519, and total output falls by 3.49 and 5.09 boxes per frame, so both terms of precision move. Tab. 1 lists all four amounts.

The observational contrast across chunk positions. Re-constructing chunk position over 20 296 labeled frames of the released evaluation and scoring every labeled box, S5-B reaches 34.445 mAP on the frames given 1–4 windows and 40.226 on those given 17–21 (Fig. 1, top). This contrast compares different frames, so it is differenced against the three RVT checkpoints, evaluated on the same frames with a state that crosses chunk boundaries. That controls for frame-position effects shared with them. It leaves 5.32 ± 0.53 points for S5-B ($z = 10.0$) and 4.37 ± 0.61 for S5-S ($z = 7.2$), against placebo differences among the RVT checkpoints of at most 0.67 points ($z = 1.56$). Permuting chunk position within each sequence puts both SSM values far outside their own null, whose standard deviation is about 0.55 points, and every placebo inside it. The two readings agree in sign and magnitude, and the paired intervention

<i>SSM-ViT, history gained by the starved block (same 3 574 frames)</i>		
	S5-B	S5-S
+0 (control)	0.00	0.00
+6 windows	3.73 ± 0.35	3.54 ± 0.39
+11 windows	4.91 ± 0.50	4.29 ± 0.52
+16 windows	5.23 ± 0.49	4.81 ± 0.52
<i>RVT, boundary moved 5 windows (same 3 574 frames)</i>		
state at boundary	RVT-s	RVT-b
carried (released)	-0.01 ± 0.03	0.05 ± 0.04
dropped	5.39 ± 0.45	6.91 ± 0.64

Table 1. Same-frame intervention on the velocity-evaluable subset: Δ mAP (points, long-history arm minus short-history arm, $\pm$ one cluster-bootstrap standard error over 406 sequences). Top: the SSM-ViT checkpoints at four boundary amounts, the last row being the intervention of Fig. 1. Bottom: RVT under the same boundary shift with its state carried and with its state dropped at every chunk start.

carries the claim because it holds the frames fixed (Supplement Sec. 1).

RVT under the same intervention. That differencing treats the RVT checkpoints as a control because their state survives a chunk boundary. The same-frame intervention applied to RVT measures that property. Driven on the SSM chunk grid, RVT reproduces its stored per-window detections bit for bit at zero shift, and the same 3 574 frames are re-scored with every boundary moved 5 windows later. With the released state carried across the boundary the paired change is -0.01 ± 0.03 points for RVT-s ($z = -0.3$) and 0.05 ± 0.04 for RVT-b ($z = 1.3$). Dropping the state at every chunk start, the condition the SSM release is already in, moves the same weights on the same frames by 5.39 ± 0.45 ($z = 12.0$) and 6.91 ± 0.64 points ($z = 10.9$). Random-frame nulls are centered within 0.112 points of zero. The boundary effect follows whether the recurrent state survives the boundary, and the control the difference-in-differences relies on holds as a measurement (Supplement Sec. 1).

What the pooled score reports. One released number pools detections whose recurrent history differs by a factor of 21. Restricting evaluation to the frames that receive 17–21 windows gives an S5-B score 1.28 points above the pooled released score, and 0.86 points for S5-S. Two systems compared under this protocol at the same nominal timestamps need not have been given the same temporal evidence, and the protocol record contains no variable by which a reader could tell. For scale, the frequency-generalization result that motivates the SSM-ViT design is a 3.31 mAP drop [23]; an unrecorded chunk boundary moves the same model by a comparable amount at a fixed frequency.

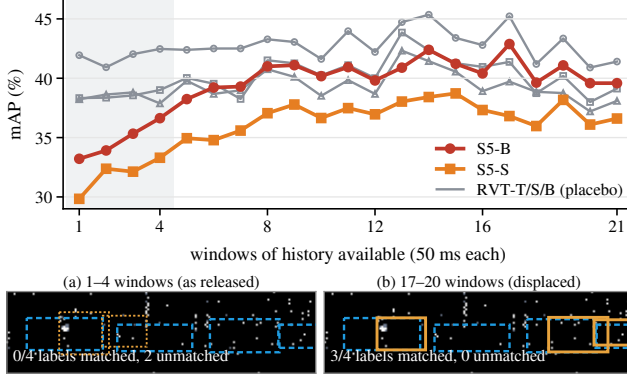


Figure 1. **Top:** benchmark score against the recurrent history the released streaming evaluation made available, at each of the 21 chunk positions, over every labeled Gen1 box with no displacement applied. Shaded: the 1–4-window block the release scores together with the rest. Between that block and frames given 17 or more windows, the two SSM checkpoints rise by 5.78 and 4.83 points; the three RVT checkpoints, scored on the same frames with state that crosses chunk boundaries, rise by at most 0.80. **Bottom:** one frame of the same-frame intervention on S5-B. Changed: the chunk boundary, and with it the recurrent history behind the detection. Fixed: frame, weights, labels (dashed), evaluator and evaluated timestamp. Solid: a detection matched to a label of its class at $\text{IoU} \geq 0.5$; dotted: unmatched; drawn above confidence 0.3 and scored at every confidence. The example is illustrative; over all 3574 frames the same intervention changes S5-B by 4.20 mAP over all labeled boxes and 5.23 ± 0.49 mAP on the velocity-evaluable subset (Supplement Sec. 1). RVT-T/S/B in the legend denotes RVT-t/s/b.

4. Supporting audits

Two audits delimit how the intervention of Sec. 3 is read. Each measures a timing descriptor that the nominal timestamp does not encode, and neither is used as a recurrent-support estimator. Their derivations, figures and tables are in Supplement Secs. 5 and 14.

Protocol record. Call a quantity *protocol-identifiable* when the protocol record of Sec. 1 determines it uniquely. The input window and the published exposure interval are in the record. The effective predictor timing, the recurrent history pooled in a score, and the attribution of event timing within an exposure are not.

Predictor side. The released RVT configuration [21] indexes each ten-bin, 50 ms input window by its end, and places that end on a label time. Ablating each bin of the released RVT-t checkpoint over 480 Gen1 validation samples puts the ablation-sensitivity centroid at -23.81 ms. A regression of detection-center residuals on label velocity, which carries the temporal calibration of [6] to a box center and is fitted on nominal-time-matched detections, gives a coefficient of -2.40 ± 7.18 ms, compatible with zero. Over a 1000 ms horizon the newest window accounts for 22.7 % of the summed window-ablation sensitivity, so the newest-

window centroid does not describe the full recurrent support (Supplement Secs. 5, 6 and 8).

Label side. Across the eighteen DSEC training sequences whose exposure files resolve [13], published exposure width varies by a factor of 127 against a frame period fixed at 50 ms. Within-box dispersion of event time inside one exposure is $206.6 \mu\text{s}$ against an analytic null of $93.0 \mu\text{s}$, so within-exposure event timing contains an illumination-locked component and cannot be attributed to object timing without separating that component first (Supplement Secs. 14 and 16). DSEC-Det ships more than one annotation version [15], and Supplement Sec. 15 names the one measured here.

5. Support-conditioned AP

Sec. 3 shows that one released protocol pools detections that received different histories. This section scores the five released Gen1 checkpoints at a stated history and compares the result with the pooled score.

Protocol. For a labeled frame scored at window r and a history length H , the detector starts from a reset recurrent state at window $r - H + 1$ and runs through window r , and the detections of window r are scored with the labels, confidence threshold, NMS and evaluator of the released-protocol dumps. We call the resulting AP at fixed H Support-Conditioned AP. H counts the windows read up to and including the scored window; we write H_{nominal} for this requested length. A frame with fewer than H windows available receives $H_{\text{forced}} = \min(H, r + 1)$ windows instead. All checkpoints are scored on the same 19946 frames, those with $H_{\text{forced}} = H_{\text{nominal}}$ at every $H \in \{1, 5, 10, 21\}$. The other 350 of the 20296 labeled frames have fewer than 21 windows of history and are excluded. The common frames lie in 406 sequences. The column *pooled* is the released evaluation of each checkpoint, in which the history depends on the release and on the frame, restricted to the same common frames. The score of the released evaluation on all 20296 frames is listed separately. The access log of every run shows that the number of distinct windows read equals H_{forced} and that the first window read is $r - H_{\text{forced}} + 1$ in every cell of every checkpoint (Supplement Sec. 1.4).

Result. Table 2 lists the scores. On the common frames the pooled ordering is RVT-b, RVT-s, S5-B, RVT-t, S5-S. At $H = 1$ the ordering is S5-B, S5-S, RVT-b, RVT-s, RVT-t, and at $H = 21$ it is S5-B, RVT-b, RVT-s, S5-S, RVT-t. Kendall's τ between the pooled ordering and the orderings at $H = 1, 5, 10$ and 21 is 0.00, 0.20, 0.20 and 0.40. At every H , each pairwise gap between S5-B and

	pooled	$H = 1$	$H = 5$	$H = 10$	$H = 21$	pooled, all frames
RVT-t	38.91 [36.94, 41.13]	27.85 [25.70, 29.99]	33.36 [31.19, 35.51]	35.15 [33.01, 37.52]	36.47 [34.34, 38.91]	38.90
RVT-s	39.70 [37.40, 42.07]	28.72 [26.57, 30.95]	34.06 [31.69, 36.23]	35.75 [33.23, 37.98]	37.02 [34.61, 39.36]	39.68
RVT-b	42.36 [39.85, 44.28]	30.25 [27.91, 32.29]	35.99 [33.78, 37.93]	37.86 [35.47, 39.86]	39.24 [36.72, 41.23]	42.32
S5-S	35.50 [33.44, 37.78]	30.48 [28.73, 32.72]	34.57 [32.65, 36.84]	35.97 [33.92, 38.17]	36.95 [34.89, 39.18]	35.55
S5-B	38.92 [36.70, 40.84]	33.69 [32.00, 35.59]	38.07 [36.03, 40.02]	39.45 [37.15, 41.42]	40.00 [37.78, 41.97]	38.95
Kendall τ against pooled	1.00	0.00	0.20	0.20	0.40	

Table 2. mAP (points) of the five released Gen1 checkpoints on the 19946 common frames, over every labeled box, with the recurrent history fixed to H windows (reset state, exactly H windows read) or as given by the released evaluation (pooled). Brackets: 95 % interval from 200 cluster-bootstrap draws over the 406 sequences, the same draw for every checkpoint and column. Last column: the released score on all 20 296 frames. The final row gives Kendall’s τ between the ordering of each column and the pooled ordering.

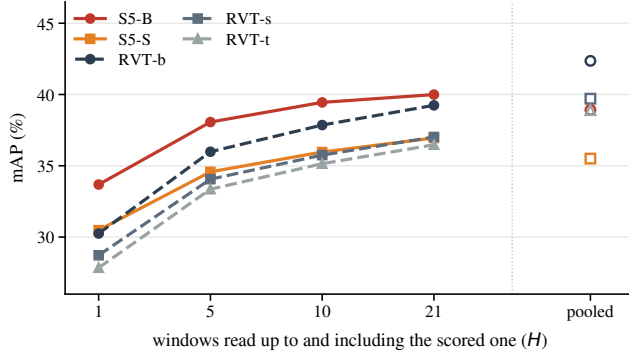


Figure 2. mAP of the five checkpoints on the common frames at $H = 1, 5, 10, 21$ (filled markers, equal spacing on the axis) and in the pooled released evaluation (open markers). RVT-t and S5-B coincide at the pooled position. Intervals are in Tab. 2 and the paired gaps in Tab. 3.

another checkpoint has a 95 % interval that excludes zero (Tab. 3), so its first rank does not depend on the bootstrap draw. The pooled ordering, by contrast, is reproduced in 48.5 % of draws on the common frames and in 42.5 % over all labeled boxes (Supplement Sec. 4).

Several pairwise gaps change sign. RVT-t minus S5-S is +3.40 [+2.58, +4.21] points in the pooled score and $-2.64 [-3.64, -1.96]$ at $H = 1$. RVT-s minus S5-S is +4.20 [+3.35, +4.88] pooled and $-1.76 [-2.66, -0.95]$ at $H = 1$. RVT-b minus S5-B is +3.44 [+2.54, +4.20] pooled and negative at every H , shrinking to $-0.75 [-1.66, -0.12]$ at $H = 21$. For RVT-s and RVT-t against S5-B the pooled intervals include zero ($+0.78 [-0.15, +1.83]$ and $-0.02 [-0.98, +1.14]$), while at every H the point gaps lie between -4.96 and -2.98 and between -5.84 and -3.52 and the intervals exclude zero. Tab. 3 lists the gaps for all pairs and Fig. 2 plots the curves over H .

What Support-Conditioned AP equalizes. The access log confirms that exactly H windows are read, and replacing the oldest admitted window changes the output in 1217 of 1218 probed cells (1216 for RVT-t), so the full admitted history reaches the detection. Every checkpoint starts from the same reset state. The protocol therefore equalizes the two things the pooled score leaves free, the reset

policy and the history length. Under the released evaluations these differ by family: the RVT release carries its state across a sequence, while the SSM-ViT release scores each chunk from a reset state (Sec. 3). This is why the pooled RVT scores sit above their $H = 21$ scores (RVT-b: 42.36 against 39.24) and the pooled S5 scores sit below theirs (S5-B: 38.92 against 40.00). The pooled column compares checkpoints that were given different support. Table 2 compares them at the same support, which is the comparison a timing benchmark is taken to make. The ordering change is a result for these five checkpoints under this evaluation and does not rank the architectures.

What a benchmark should record. A nominal timestamp does not determine the effective temporal support or the label-acquisition support, and the intervention above shows a case in which the unrecorded support changes the score. We propose that a benchmark record, for each scored sample, five fields. The *nominal timestamp* is the instant at which the sample is scored. The *input window* is what the detector reads at that timestamp. The *state-reset policy* states where the recurrent state is cleared, including chunk boundaries and sequence starts. The *effective history length* is the number of windows read, as the access log of the evaluation run records it. The *label-acquisition support* is the interval over which the label’s evidence was acquired. Together, the reset policy and the history length determine the effective temporal support. Tab. 4 fills in these fields for the two releases examined here. A score is then reported at a declared history length, as in Tab. 2, together with the distribution of history lengths that the pooled score contains. These fields do not equalize the support across checkpoints. They record it, so that scores can be compared at a history that both detectors received.

6. Conclusion

With frames, weights, labels, evaluator and evaluated timestamps fixed, moving only the chunk boundary shifts S5-B by 4.20 mAP over all labeled validation boxes and by 5.23 mAP on the velocity-evaluable subset. The chunk boundary is a degree of freedom the protocol does not record. It

pair (first – second)	pooled	$H = 1$	$H = 5$	$H = 10$	$H = 21$
RVT-t – RVT-s	-0.80 [-1.42, -0.14]	-0.88 [-1.54, -0.31]	-0.70 [-1.26, -0.17]	-0.60 [-1.14, -0.04]	-0.55 [-1.06, -0.05]
RVT-t – RVT-b	-3.45 [-4.21, -2.35]	-2.40 [-2.86, -1.83]	-2.63 [-3.15, -2.11]	-2.71 [-3.18, -2.03]	-2.77 [-3.40, -1.88]
RVT-t – S5-S	+3.40 [+2.58, +4.21]	-2.64 [-3.64, -1.96]	-1.22 [-2.23, -0.56]	-0.82 [-1.72, -0.30]	-0.48 [-1.37, +0.09]
RVT-t – S5-B	-0.02 [-0.98, +1.14]	-5.84 [-6.92, -5.05]	-4.71 [-5.89, -3.75]	-4.30 [-5.37, -3.47]	-3.52 [-4.38, -2.54]
RVT-s – RVT-b	-2.65 [-3.15, -1.92]	-1.53 [-1.91, -0.85]	-1.93 [-2.51, -1.27]	-2.11 [-2.77, -1.49]	-2.22 [-2.80, -1.44]
RVT-s – S5-S	+4.20 [+3.35, +4.88]	-1.76 [-2.66, -0.95]	-0.51 [-1.56, +0.38]	-0.22 [-1.22, +0.60]	+0.07 [-0.86, +0.78]
RVT-s – S5-B	+0.78 [-0.15, +1.83]	-4.96 [-6.15, -3.90]	-4.01 [-5.23, -2.93]	-3.70 [-5.00, -2.75]	-2.98 [-4.08, -1.98]
RVT-b – S5-S	+6.85 [+5.73, +7.61]	-0.23 [-1.23, +0.36]	+1.41 [+0.49, +1.94]	+1.89 [+0.90, +2.42]	+2.29 [+1.31, +2.82]
RVT-b – S5-B	+3.44 [+2.54, +4.20]	-3.44 [-4.54, -2.76]	-2.08 [-3.16, -1.41]	-1.59 [-2.52, -0.96]	-0.75 [-1.66, -0.12]
S5-S – S5-B	-3.42 [-4.03, -2.79]	-3.20 [-3.83, -2.70]	-3.49 [-4.09, -2.83]	-3.48 [-4.08, -2.81]	-3.04 [-3.59, -2.50]

Table 3. Paired mAP gaps (points) between the five released Gen1 checkpoints on the 19 946 common frames, with 95 % intervals from 200 bootstrap draws over the 406 sequences. Pooled is the released evaluation restricted to the same frames. Gaps are computed from unrounded scores. A gap whose interval excludes zero is a difference the sequence resampling does not remove. The sign of the pooled gap and the sign at $H = 1$ are opposite, with both intervals excluding zero, for RVT-t – S5-S, RVT-s – S5-S and RVT-b – S5-B.

field	RVT release	SSM-ViT release
nominal timestamp	recorded	recorded
input window	recorded	recorded
state-reset policy	sequence start only; not recorded	every 21-window chunk; not recorded
effective history length	up to sequence length; not recorded	1–21 windows; not recorded

Table 4. Four of the five fields of Sec. 5 for the two released Gen1 evaluations. A field marked not recorded is not part of what the protocol reports with a score. The fifth field, label-acquisition support, belongs to the dataset and is audited on DSEC in Sec. 4.

changes the recurrent support behind a reported score while every recorded quantity stays fixed, which is an instance where temporal-support identifiability does not hold. RVT shows the same effect as soon as its state is reset at the boundary, so the effect belongs to the reset policy and not to one architecture.

Support-Conditioned AP removes this degree of freedom by fixing the reset state and the history behind every score. On five released Gen1 checkpoints, S5-B moves from third in the pooled score to first at every H , Kendall’s τ between the pooled and fixed- H orderings stays at or below 0.40, and three pairwise gaps reverse sign with intervals that exclude zero on both sides.

Two scores at the same nominal timestamp are comparable in time only when the support behind them is the same. A benchmark can make this checkable by recording five fields for each scored sample (nominal timestamp, input window, state-reset policy, effective history length, label-acquisition support) and by reporting scores at a declared history length.

References

- [1] W.-C. Hung, V. Casser, H. Kretschmar, J.-J. Hwang, and D. Anguelov. LET-3D-AP: Longitudinal error tolerant 3D average precision for camera-only 3D detection. *arXiv:2206.07705*, 2022. 2
- [2] A. Kugele, T. Pfeil, M. Pfeiffer, and E. Chicca. How many events make an object? Improving single-frame object detection on the 1 Mpx dataset. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2023, pages 3913–3922. 2
- [3] Waymo Open Dataset. `label.proto`, `field camera_synced_box` and its documentation comment. 2
- [4] P. Sun et al. Scalability in perception for autonomous driving: Waymo Open Dataset. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [5] A. J. Miro, L. af Klinteberg, B. Timus, A. Asefaw, A. Khoche, T. Gustafsson, S. S. Mansouri, and M. Daneshmand. Correcting and quantifying systematic errors in 3D box annotations for autonomous driving. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026. 2
- [6] T. Qin and S. Shen. Online temporal calibration for monocular visual-inertial systems. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018. 4
- [7] M. Li, Y.-X. Wang, and D. Ramanan. Towards streaming perception. In *European Conference on Computer Vision (ECCV)*, 2020. 2
- [8] H. Mao, X. Yang, and W. J. Dally. A delay metric for video object detection: What average precision fails to tell. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pages 573–582. 2
- [9] D. Bolya, S. Foley, J. Hays, and J. Hoffman. TIDE: A general toolbox for identifying object detection errors. In *European Conference on Computer Vision (ECCV)*, 2020. 2
- [10] D. Hoiem, Y. Chodpathumwan, and Q. Dai. Diagnosing error in object detectors. In *European Conference on Computer Vision (ECCV)*, 2012.
- [11] H. Alwassel, F. Caba Heilbron, V. Escorcia, and B. Ghanem. Diagnosing error in temporal action detectors. In *European Conference on Computer Vision (ECCV)*, 2018.
- [12] J. Luiten, A. Ošep, P. Dendorfer, P. Torr, A. Geiger, L. Leal-Taixé, and B. Leibe. HOTA: A higher order metric for evaluating multi-object tracking. *International Journal of Computer Vision*, 2021. 2
- [13] M. Gehrig, W. Aarents, D. Gehrig, and D. Scaramuzza. DSEC: A stereo event camera dataset for driving scenarios. *IEEE Robotics and Automation Letters*, 2021. 4
- [14] D. Gehrig and D. Scaramuzza. Low-latency automotive vision with event cameras. *Nature*, 629:1034–1040, 2024. 2
- [15] K. Wang, S. Liu, H. Shi, L. Shi, and H. Chen. Beyond duality: A hybrid framework of leveraging shared and private features for RGB-Event object detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. 4
- [16] P. de Tournemire, D. Nitti, E. Perot, D. Migliore, and A. Sironi. A large scale event-based detection dataset for automotive. *arXiv:2001.08499*, 2020. 2
- [17] E. Perot, P. de Tournemire, D. Nitti, J. Masci, and A. Sironi. Learning to detect objects with a 1 megapixel event camera. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [18] H. Cho, J.-Y. Kang, Y. Kim, and K.-J. Yoon. Ev-3DOD: Pushing the temporal boundaries of 3D object detection with event cameras. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 2
- [19] L. Sun, C. Sakaridis, J. Liang, Q. Jiang, K. Yang, P. Sun, Y. Ye, K. Wang, and L. Van Gool. Event-based fusion for motion deblurring with cross-modal attention. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 2
- [20] T. Kim, J. Lee, L. Wang, and K.-J. Yoon. Event-guided deblurring of unknown exposure time videos. In *European Conference on Computer Vision (ECCV)*, 2022. 2
- [21] M. Gehrig and D. Scaramuzza. Recurrent vision transformers for object detection with event cameras. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 2, 4
- [22] H. Liu, S. Peng, L. Zhu, Y. Chang, H. Zhou, and L. Yan. Seeing Motion at Nighttime with an Event Camera. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2
- [23] N. Zubić, M. Gehrig, and D. Scaramuzza. State space models for event cameras. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2, 3
- [24] Z. Wu, M. Gehrig, Q. Lyu, X. Liu, and I. Gilitschenski. LEOD: Label-efficient object detection for event cameras. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16933–16943, 2024. 2
- [25] Z. Sui, H. Hao, W. He, S.-H. Lee, and W. Wang. Adaptive spatial-temporal window: unlocking the potential of event cameras in heterogeneous velocity scenarios. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 946–955, 2026. 2
- [26] Y. Yang, J. Liang, B. Yu, Y. Chen, J. S. Ren, and B. Shi. Latency correction for event-guided deblurring and frame interpolation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 24977–24986, 2024. 2
- [27] J. Chu, R. Zhang, C. Yang, Z. Yu, Z. Bu, H. Liu, F. Röhrbein, A. Knoll, G. Chen, and C. Jiang. Bridging the latency gap with a continuous stream evaluation framework in event-driven perception. *Nature Communications*, 17:2441, 2026. 2
- [28] S. Kuzucu, K. Oksuz, J. Sadeghi, and P. K. Dokania. On calibration of object detectors: Pitfalls, evaluation and baselines. In *European Conference on Computer Vision (ECCV)*, 2024. 2
- [29] R. Fogliato, P. Patil, M. Monfort, and P. Perona. A framework for efficient model evaluation through stratification, sampling, and estimation. In *European Conference on Computer Vision (ECCV)*, 2024.
- [30] C. G. Northcutt, A. Athalye, and J. Mueller. Pervasive label errors in test sets destabilize machine learning benchmarks. In *NeurIPS Datasets and Benchmarks*, 2021.
- [31] B. Ercan, O. Eker, A. Erdem, and E. Erdem. EVREAL: Towards a comprehensive benchmark and analysis suite for event-based video reconstruction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2023. 2
- [32] J. Yang, S. Liu, Z. Li, X. Li, and J. Sun. Real-time object detection for streaming perception. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. 2
- [33] R. Salles et al. SoftED: Metrics for soft evaluation of time series event detection. *Computers & Industrial Engineering*, 2024. 2
- [34] Haitian Zhang et al. Frequency-adaptive low-latency object detection using events and frames. *arXiv:2412.04149*, 2024. 2